STAT 7900: Special Topic: Causal Analysis



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STAT 7900

Before Any Estimator: Define the Causal Target

$$\mathrm{ATE} = \mathrm{E}[\mathrm{Y}(1) - \mathrm{Y}(0)]$$

Population

Whose effect do we want?

Treatment & comparator

What intervention versus what alternative?

Outcome & time

What endpoint, measured when?

Estimator choice comes after the estimand and identification assumptions.

The Common Identification Assumptions

Consistency

Observed Y equals the potential outcome under the treatment actually received.

Exchangeability

$$Y(t) \perp T \mid X$$

Positivity

Both treatments are possible within relevant X profiles.

No interference

One unit's treatment does not alter another unit's outcome.

These methods estimate causal effects only after identification is justified.

Recap: Four Ways to Estimate Causal Effects

g-computation

Model outcomes and predict both treatment worlds.

PSM

Select comparable observed treatment and control units.

IPW

Reweight observed units to construct a pseudo-population.

AIPW

Combine outcome predictions with weighting corrections.

Different estimators • same causal identification problem

One Causal Question: Four Estimation Strategies

g-computation

- Use X to model Y.
- Predict Y(1) and Y(0).
- Average the contrast.

PSM

- Use X to model T.
- Find comparable units.
- Compare matched outcomes.

IPW

- Use X to model T.
- Create inverse-probability weights.
- Compare weighted outcomes.

AIPW

- Model both Y and T.
- Predict outcomes.
- Add weighted residual corrections.

The key distinction: model outcomes, redesign the sample, reweight the sample, or combine models.

g-Formula: Identification Comes First

$$E[Y(t)] = E_X\{\, E[Y \mid T = t, X]\,\}$$

Within X

Use comparable observed units to estimate the treatment-specific mean.

Intervene conceptually

Set $T = t$ while keeping each person's baseline X fixed.

Standardize

Average over one common target distribution of X.

The DAG and assumptions justify replacing missing counterfactual means with observed-data quantities.

g-Computation: Predict Two Worlds for Everyone

Step 1

Fit

$$m(t,x) = E[Y|T = t, X = x].$$

Step 2

Predict $\hat{m}(1, X_i)$.

Predict $\hat{m}(0, X_i)$.

Step 3

Average

$$\hat{m}(1, X_i) - \hat{m}(0, X_i).$$

$$\hat{\tau}_g = n^{-1}\Sigma_i \{\hat{m}(1, X_i) - \hat{m}(0, X_i)\}$$

g-Computation in R: The Logic

Code

```
fit <- lm(re78 ~ treat + age + educ + race + married +
          nodegree + re74 + re75, data = d)

y1 <- predict(fit, transform(d, treat = 1))
y0 <- predict(fit, transform(d, treat = 0))
mean(y1 - y0)
```

What each line means

Fit an outcome model.

Create the all-treated prediction world.

Create the all-control prediction world.

Average the individual predicted contrasts.

Main vulnerability: outcome-model misspecification.

In an additive linear model without T×X interaction, the g-computation ATE equals the treatment coefficient.

PSM: Construct Comparison Before Outcomes

$$e(X) = P(T = 1 \mid X)$$

Estimate $e(X)$

Summarize treatment assignment using baseline covariates.

Match

Find treated and control units with similar scores.

Check balance

The matched groups should be similar on measured baseline X.

Matching uses observed comparison people to support the missing counterfactual outcome.

Nearest Neighbor, Calipers, and Target Population

Nearest neighbor

Choose the control with the closest propensity score.

Caliper

Reject matches that are still too far apart.

Estimand

Treated-to-control matching often targets ATT or the effect among retained treated units.

A better match can mean a smaller, but more defensible, target population.

Matching Is Not Finished When the Algorithm Stops

Balance

Check standardized mean differences on baseline covariates.

Overlap / retention

How many units were discarded? Where is common support?

Inference

Respect matched pairs, replacement, and unequal matching weights.

A small propensity-score gap is only a starting point; covariate balance determines usefulness.

IPW: Keep Units, Change Their Contribution

ATE weights: treated - $1/e(X)$ • control - $1/\{1 - e(X)\}$

Rare observed assignment

Gets a larger weight.

Common observed assignment

Gets a smaller weight.

Goal

Create a weighted pseudo-population with comparable baseline X.

Think survey sampling: low-probability observations represent more of the target population.

ATE and ATT Weights Target Different Populations

ATE weights

$$T = 1: \quad 1/e(X)$$
$$T = 0: \quad 1/\{1 - e(X)\}$$

Target: overall study population.

ATT weights

$$T = 1: \quad 1$$
$$T = 0: \quad e(X)/\{1 - e(X)\}$$

Target: treated population.

Same data + same propensity model + different weights = different causal question.

IPW Diagnostics: Do the Weights Behave?

Extreme weights

- $e(X) \approx 0$ for treated or $e(X) \approx 1$ for controls can create huge weights.
- Inspect maxima and upper quantiles.

Effective sample size

- ESS = $(\sum w)^2 \,/\, \Sigma w^2$
- Large weight concentration lowers ESS.

Balance after weighting

- Recalculate SMDs using weights.
- Good code execution is not enough.

Stabilization rescales weights; it does not repair poor overlap.

AIPW Formula: Read It in Pieces

$$\hat{\tau}_{AIPW} = n^{-1} \sum_i [\, \hat{m}_1(X_i) - \hat{m}_0(X_i)] + T_i \frac{\{Y_i - \hat{m}_1(X_i)\}}{\hat{e}(X_i)} - (1 - T_i)\{Y_i - \hat{m}_0(X_i)\}/\{1 - \hat{e}(X_i)\}\,]$$

Prediction contrast

What the outcome model says.

Treated correction

Observed treated residuals, reweighted.

Control correction

Observed control residuals, reweighted.

No need to memorize the entire expression: remember prediction + two residual corrections.

Double Robustness: What It Means

What it means

Under standard conditions, AIPW remains consistent if:

- the outcome model is correct, OR
- the propensity model is correct.

What it does NOT mean

It does not fix:

- unmeasured confounding
- lack of positivity
- two badly misspecified nuisance models

Double robustness protects against one nuisance-model misspecification—not every causal failure.

Same Job-Training Data: Different Answers

g-computation

ATE ≈ +$1,548

Outcome-model based

PSM

0.05-caliper match ≈ +$1,433

113 retained treated

ATT-like target

Normalized IPW

ATE ≈ +$225

Propensity-model based

AIPW

ATE ≈ +$887

Outcome + propensity models

Do not compare the numbers blindly: PSM here targets a retained-treated population, whereas the other three shown target the overall study-population ATE.

One-Slide Comparison

Feature	g-computation	PSM	IPW	AIPW
Main model	Outcome model	Propensity model	Propensity model	Outcome + propensity
How comparison is built	Predict both worlds	Select comparable units	Reweight units	Predict + weighted correction
Keeps all units?	Usually yes	Often no	Usually yes	Usually yes
Typical target	ATE / ATT by standardization	Often ATT-like	ATE or ATT	ATE or ATT
Main diagnostic	Outcome-model adequacy	Balance + overlap	Weights + ESS + balance	Both model & weight diagnostics
Main vulnerability	Outcome misspecification	Poor matches / support	Extreme weights / PS misspecification	Both models wrong; poor overlap

How Should I Choose Among Them?

Prefer g-computation when

You can model the outcome well.

You want flexible standardization.

Interactions / nonlinear response surfaces are scientifically central.

Prefer PSM / IPW when

Design-stage comparability is central.

You want explicit balance diagnostics.

You can defend a useful treatment-assignment model.

Prefer AIPW when

You want both nuisance models.

You can diagnose both sides.

You may use flexible nuisance estimators with cross-fitting.

There is no automatic winner: estimand, overlap, data structure, and modeling credibility matter.

STAT 7900 - Quiz 6



From ATE to Heterogeneous Treatment Effects

So far

Most of our estimators have targeted one average effect.

Now

Ask whether the effect changes across baseline characteristics X.

Goal

Move from ATE to conditional average treatment effects.

$$\text{ATE} \rightarrow \text{CATE}(x) \rightarrow \text{treatment-effect heterogeneity}$$

Why Average Treatment Effect Can Hide the Story

Group A

Treatment effect
= +10

Group B

Treatment effect
= 0

Group C

Treatment effect
= −10

A single overall average can conceal benefit, no effect, and harm.

A small or zero ATE does not imply that treatment is unimportant for everyone.

ATE vs CATE

$$ATE = E[Y(1) - Y(0)]$$

$$CATE(x) = E[Y(1) - Y(0) \mid X = x]$$

ATE

How much does treatment help in the target population, on average?

CATE

How much does treatment help for people with a particular X profile?

Possible X

Age • severity • biomarker • baseline risk • prior outcome

ATE averages over X. CATE conditions on X.

A small or zero ATE does not imply that treatment is unimportant for everyone.

CATE Is Still an Average

Individual treatment effect

$$Y_i(1) - Y_i(0)$$

Defined for one specific person.

We never observe both potential outcomes for that person.

$$CATE(x)$$

Average effect among people with X=x.

Still a population-level quantity.

More local than the ATE, but not a deterministic prediction for one person.

"Patients like this benefit by 8 units on average" ≠ "this patient will benefit by exactly 8."

Effect Modifier vs Confounder

Confounder

Affects treatment assignment and outcome.

Creates a backdoor path.

Threatens identification.

Question: what must we adjust for?

Effect modifier

Changes the magnitude or direction of the treatment effect.

Does not have to cause treatment.

Question: for whom is the effect different?

Confounding is about bias. Effect modification is about genuine variation in the causal effect.

Effect Modifier vs Confounder

Example: Age

Older patients may be more likely to receive treatment.

Confounding role

Age may also predict the outcome even without treatment.

Effect-modification role

The biological treatment response may also differ by age.

One X can matter for identification AND for heterogeneity.

Adjusting for age removes confounding; an interaction with age investigates whether the effect itself varies.

Real-World Example

Low-risk patients

Control risk: 4%

Treatment risk: 3%

Risk difference: −1 percentage point

Medium-risk patients

Control risk: 15%

Treatment risk: 10%

Risk difference: −5 percentage points

High-risk patients

Control risk: 35%

Treatment risk: 20%

Risk difference: −15 percentage points

Same treatment, but much larger absolute benefit in higher-risk patients.

Real-World Heterogeneity Across Fields

Medicine

- Cancer therapy by biomarker status
- Drug benefit by baseline cardiovascular risk
- Benefit-to-harm balance by frailty

Education

- Tutoring by prior achievement
- Class-size reduction by baseline performance
- Technology interventions by access

Policy

- Job training by prior earnings
- Cash transfers by baseline income
- Policy effects by region

HTE turns “Does it work?” into “Where, for whom, and how much?”

What Counts as Evidence of Heterogeneity?

Not enough

Treatment is significant in one subgroup but not another.

Two subgroup point estimates look visually different.

Better evidence

Directly test an interaction.

Estimate the difference in treatment effects.

Report uncertainty for that difference.

Best practice

Use pre-specified modifiers when possible.

Avoid data-driven subgroup fishing.

Validate exploratory findings.

The question is whether treatment effects differ, not whether two separate p-values differ.

Next: Model the Treatment Effect as a Function of X

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]$$

Classical approach

Treatment × covariate interactions

Flexible approach

S-, T-, and X-learners

Modern approach

Causal forests and data-adaptive CATE estimation

Next section: Classical THE, subgroup effects and interaction models.

Classical HTE: Interaction Models

Average-effect model

One treatment coefficient for everyone.

Interaction model

Allow treatment effect to depend on X.

Target

Estimate and interpret $\tau(x)$.

$$\text{one effect} \rightarrow \text{treatment} \times \text{covariate interaction} \rightarrow \text{effect curve}$$

The Basic Interaction Model

$$E[Y \mid T, X] = \beta_0 + \beta_1 T + \beta_2 X + \beta_3 (T \times X)$$

β_0

Mean outcome when $T = 0$ and $X = 0$.

β_1

Treatment effect when $X = 0$.

β_2

Association of X with outcome when $T = 0$.

β_3

How the treatment effect changes as X changes.

The interaction coefficient β_3 is the key parameter for classical HTE.

Deriving the Conditional Treatment Effect

$$m(1, x) = \beta_0 + \beta_1 + \beta_2 x + \beta_3 x$$

$$m(0, x) = \beta_0 + \beta_2 x$$

$$\tau(x) = m(1, x) - m(0, x) = \beta_1 + \beta_3 x$$

β_1

Treatment effect at the reference value $X = 0$.

β_3

Change in treatment effect for a one-unit increase in X.

What Does β_1 Mean? The Reference Value Matters

Suppose $X = age$

If age is entered directly, $X = 0$ means age 0.

Then β_1 is the treatment effect at age 0.

That may be mathematically valid but scientifically useless.

Better strategy

Center age at a meaningful value.

For example: $X_c = age - 50$.

Then β_1 becomes the treatment effect at age 50.

Main lesson: with an interaction, the 'treatment coefficient' is conditional on the coding of X.

What Does β_3 Mean?

$$\tau(x+1) - \tau(x) = \beta_3$$

$\beta_3 > 0$

Treatment benefit becomes more positive as X increases.

$\beta_3 = 0$

No evidence of linear effect modification by X.

$\beta_3 < 0$

Treatment effect becomes less positive or more harmful as X increases.

β_3 describes a difference in effects, not merely a difference in outcomes.

Binary Effect Modifier: Diabetes Status

$$E[Y|T,D] = \beta_0 + \beta_1 T + \beta_2 D + \beta_3 (T \times D)$$

D = 0: no diabetes

Treatment effect = β_1

Example: $\beta_1 = -3$ mmHg

$D = 1$: diabetes

Treatment effect = $\beta_1 + \beta_3$

Example: $\beta_3 = -5 \rightarrow$ effect = -8 mmHg

The interaction asks whether the treatment effect differs by diabetes status.

Worked Example: Read the Interaction

Group	Control mean	Treatment mean	Effect
No diabetes	130	127	−3
Diabetes	145	137	−8

$$\text{Difference in treatment effects} = (-8) - (-3) = -5$$

That difference corresponds to the interaction coefficient β_3.

Continuous Effect Modifier: Baseline Severity

$$\tau(x) = \beta_1 + \beta_3 x$$

Example coefficients

$\beta_1 = -2$

$\beta_3 = -1.5$

At X = 0

$\tau(0) = -2$

At X = 4

$\tau(4)$
$= -2 - 1.5 \times 4$
$= -8$

A negative β_3 means the treatment becomes increasingly beneficial as severity increases, when lower Y is better.

Treatment Effect as a Function

$X = 0$

$\tau(0) = -2$

$X = 1$

$\tau(1) = -3.5$

$X = 2$

$\tau(2) = -5$

$X = 4$

$\tau(4) = -8$

For a continuous modifier, avoid forcing everyone into arbitrary 'low' and 'high' groups.

Centering a Continuous Modifier

$$X_c = X - 50$$

At age 50

$X_c = 0$, so β_1 is the treatment effect at age 50.

At age 60

$X_c = 10$, so $\tau(60) = \beta_1 + 10\beta_3$.

Why this helps

Main-effect coefficients become interpretable at a meaningful reference point.

Centering changes interpretation, not the fitted values or evidence for interaction.

R: Fit and Read an Interaction Model

Code

```
d$age50 <- d$age - 50
fit <- lm(Y ~ treat * age50 + sex + baseline_risk, data = d)
summary(fit)
coef(fit)[c("treat", "treat:age50")]
```

Interpretation

treat = estimated effect at age 50.

treat: age50 = change in treatment effect per one-year increase in age.

Other baseline covariates can still be included for confounding adjustment / precision.

Do not interpret the interaction model by looking only at the p-value for treat.

Uncertainty for $\tau(x)$

$$\hat{\tau}(x) = \hat{\beta}_1 + x\,\hat{\beta}_3$$

$$Var\{\hat{\tau}(x)\} = Var(\hat{\beta}_1) + x^2 Var(\hat{\beta}_3) + 2xCov(\hat{\beta}_1, \hat{\beta}_3)$$

Why covariance matters

$\hat{\beta}_1$ and $\hat{\beta}_3$ are estimated together; their uncertainty is not independent.

What to report

$\hat{\tau}(x)$ with a confidence interval at scientifically relevant X values.

What to plot

Effect curve plus uncertainty band across the supported range of X.

Do not extrapolate the interaction effect far beyond the observed range of X.

A Mistake: 'Significant Here, Not Significant There'

Subgroup A

Treatment effect = −6
95% CI: [−10, −2]
p = 0.01

Subgroup B

Treatment effect = −4
95% CI: [−9, 1]
p = 0.11

Can we conclude the effects differ? NO.

The correct question is whether (effect A − effect B) differs from zero.

Correct HTE Test, and the Limit of One Interaction

Correct classical test

Fit one model containing $T \times X$.

Test H_0: $\beta_3 = 0$.

Estimate the difference in treatment effects directly.

Report $\tau(x)$, not only the interaction p-value.

Where this becomes difficult

Many possible modifiers.

Nonlinear treatment-effect patterns.

Higher-order interactions.

Data-driven discovery without massive subgroup fishing.

Next: flexible CATE estimation with S-, T-, X-learners and causal forests.

Modern CATE Estimation

Classical HTE

Choose X first, then specify $T \times X$ interactions.

Modern HTE

Let flexible prediction models learn complex response surfaces.

Goal

Estimate τ(x) without assuming a simple linear interaction.

Outcome learning → counterfactual prediction → CATE estimation

Why Not Just Add More Interactions?

Many modifiers

- Age
- Biomarkers
- Baseline risk
- Comorbidities
- Prior outcomes

Complex shapes

- Nonlinear effects
- Thresholds
- Higher-order interactions
- Unknown functional form

Statistical cost

- Many parameters
- Low power
- Overfitting
- Multiplicity

Modern HTE methods trade rigid parametric structure for flexible nuisance and CATE estimation.

Meta-Learners: Use Prediction Algorithms

Base learner

Regression, random forest, boosting, neural network, etc.

Meta-learner

A recipe for organizing one or more prediction models around treatment and outcome.

Causal target

Convert those predictions into an estimate of $\tau(x)$.

The S-, T-, and X-learners are strategies, not specific machine-learning algorithms.

S-Learner: One Outcome Model

$$m(t, x) = E[Y \mid T = t, X = x]$$

Fit once

Use T and X together in one flexible prediction model.

Predict twice

Set $T = 1 \rightarrow \hat{m}(1, x)$

Set $T = 0 \rightarrow \hat{m}(0, x)$

Subtract

$\hat{\tau}_S(x)$
$= \hat{m}(1, x) - \hat{m}(0, x)$

S = single model.

S-Learner: Strengths and Weaknesses

Strengths

- Simple workflow
- Shares information across treatment arms
- Can use powerful nonlinear learners
- Works naturally with many X

Weaknesses

- Treatment may be ignored if X predicts Y strongly
- Regularization can shrink treatment interactions
- One model may underrepresent true arm-specific structure

Good outcome prediction does not automatically imply good CATE estimation.

R Logic for an S-Learner

Conceptual code

```
fit_s <- learner(Y ~ treat + X1 + X2 +
..., data=d)

d1 <- transform(d, treat=1)
d0 <- transform(d, treat=0)

tau_s <- predict(fit_s, d1) -
predict(fit_s, d0)
```

What matters

- The learner must allow treatment to interact with X somehow.
- A plain additive linear model produces no heterogeneity unless interactions are included.
- Flexible learners can learn interactions implicitly.

Prediction under two treatment worlds is the same g-computation idea, now used locally for τ(x).

T-Learner: Two Outcome Models

$$m_1(x) = E[Y|T = 1, X = x] \quad \text{and} \quad m_0(x) = E[Y|T = 0, X = x]$$

Treated model

Fit a model using only treated units.

Control model

Fit a separate model using only control units.

Subtract

$$\hat{\tau}_T(x) = \hat{m}_1(x) - \hat{m}_0(x)$$

T = two models.

T-Learner: When Separate Models Help, and Hurt

Why it can help

- Treatment and control response surfaces may be very different.
- Each arm can learn its own nonlinearities.
- No need to encode treatment interactions explicitly.

Why it can hurt

- Each model sees fewer observations.
- Severe treatment imbalance can make one model unstable.
- Errors from two separate models accumulate in $\tau(x)$.

The T-learner favors flexibility across arms but sacrifices information sharing.

S-Learner vs T-Learner: The Key Tradeoff

S-Learner

- 1 model
- More information sharing
- Potentially smoother / more stable
- May shrink heterogeneity toward zero

T-Learner

- 2 models
- Less information sharing
- Allows very different response surfaces
- Can be unstable in small treatment arms

Bias–variance tradeoff appears in CATE estimation too.

X-Learner: Imputed Treatment Contrasts

1. Fit outcomes

Estimate $\hat{m}_1(x)$ and $\hat{m}_0(x)$.

2. Impute effects

Construct a pseudo-effect for each observed unit.

3. Model effects

Learn treatment-effect patterns as functions of X.

4. Combine

Blend arm-specific effect models.

The X-learner explicitly creates proxy outcomes for the missing treatment effect.

X-Learner: The Four Steps

For treated: $D^1 = Y - \hat{m}_0(X)$

For controls: $D^0 = \hat{m}_1(X) - Y$

Then model D^1

Estimate $\hat{\tau}_1(x)$ among treated units.

$$\hat{\tau}_1(x) \approx \mathrm{E}\left[D^1 | X = x, T = 1\right]$$

Model D^0

Estimate $\hat{\tau}_0(x)$ among control units.

$$\hat{\tau}_0(x) \approx \mathrm{E}\left[D^0 | X = x, T = 0\right]$$

Combine

$$\hat{\tau}_X(x) = g(x)\hat{\tau}_0(x) + \{1 - g(x)\}\hat{\tau}_1(x)$$

The weighting function $g(x)$ often uses treatment propensity information.

When Is the X-Learner Attractive?

Treatment imbalance

One arm may be much larger than the other.

Structured heterogeneity

Treatment effects may be simpler than outcome surfaces.

Caution

Still depends on accurate nuisance learning and adequate local overlap.

X-learning can borrow strength strategically; it cannot manufacture missing support.

Comparison: S-, T-, and X-Learners

Feature	S-Learner	T-Learner	X-Learner
Outcome models	1	2	2 initial + effect models
Information sharing	High	Low	Intermediate / adaptive
Treatment imbalance	Can be stable	Can struggle	Often attractive
Main idea	Predict both worlds from one model	Separate response surfaces	Impute and learn treatment effects
Main risk	Treatment signal shrunk	Small-arm instability	More modeling stages

None is uniformly best; performance depends on sample size, imbalance, smoothness, and the base learner.

R Code: S-Learner

Code

```
fit_s <- lm(
  re78 ~ treat * (age + educ + re74 + re75),
  data = d
)
d1 <- transform(d, treat = 1)
d0 <- transform(d, treat = 0)
mu1_s <- predict(fit_s, newdata = d1)
mu0_s <- predict(fit_s, newdata = d0)
tau_s <- mu1_s - mu0_s
summary(tau_s)
```

How to read it

1. Fit one outcome model using T and X together.
2. Create the all-treated and all-control prediction worlds.
3. For each person, subtract the two predicted outcomes.

Important: with a linear model, include $treat \times X$ if you want heterogeneity.

R Code: T-Learner

Code

```
fit1 <- lm(
  re78 ~ age + educ + re74 + re75,
  data = d[d$treat == 1, ]
)
fit0 <- lm(
  re78 ~ age + educ + re74 + re75,
  data = d[d$treat == 0, ]
)
mu1_t <- predict(fit1, newdata = d)
mu0_t <- predict(fit0, newdata = d)
tau_t <- mu1_t - mu0_t
summary(tau_t)
```

How to read it

1. Fit a treated-outcome model only among treated units.
2. Fit a control-outcome model only among control units.
3. Use both models to predict for the same person.
4. Subtract the two predicted outcomes.

This allows the treated and control response surfaces to differ.

R Code: X-Learner (Steps 1–2)

Code

```
# Step 1: initial outcome models
fit1 <- lm(re78 ~ age + educ + re74 + re75,
           data = d[d$treat == 1, ])
fit0 <- lm(re78 ~ age + educ + re74 + re75,
           data = d[d$treat == 0, ])

m1 <- predict(fit1, newdata = d)
m0 <- predict(fit0, newdata = d)

# Step 2: imputed treatment effects
D1 <- d$re78[d$treat == 1] - m0[d$treat == 1]
D0 <- m1[d$treat == 0] - d$re78[d$treat == 0]
```

How to read it

For treated units:

D1 = observed Y(1) − predicted Y(0)

For control units:

D0 = predicted Y(1) − observed Y(0)

These are pseudo-treatment effects.

R Code: X-Learner (Steps 3–4)

Code

```
# Step 3: model the pseudo-effects
fit_tau1 <- lm(D1 ~ age + educ + re74 + re75,
               data = d[d$treat == 1, ])
fit_tau0 <- lm(D0 ~ age + educ + re74 + re75,
               data = d[d$treat == 0, ])
tau1 <- predict(fit_tau1, newdata = d)
tau0 <- predict(fit_tau0, newdata = d)

# Step 4: combine
ps_fit <- glm(treat ~ age + educ + re74 +
re75,
              family = binomial(), data = d)
e <- predict(ps_fit, type = 'response')
tau_x <- e * tau0 + (1 - e) * tau1
summary(tau_x)
```

How to read it

1. Learn how the pseudo-effects vary with X.
2. Predict those effect models for everyone.
3. Combine the two effect-model predictions.

The propensity score helps decide how much weight to give each side.

Why Do We Need Another HTE Method?

S / T / X learners

Learn outcome models and/or imputed effects.

Result depends on the choice and quality of base learners.

What is still difficult?

Many modifiers, nonlinear interactions, sparse subgroups, and data-driven searches for where effects differ.

A different idea

Build local treatment–control comparisons around people with similar X, using trees designed for HTE.

Causal forest is an alternative CATE estimator, not a required successor to S/T/X.

Ordinary vs Causal Trees: What Should a Split Find?

Ordinary prediction tree

Split on X to make outcomes Y easier to predict.

Example: baseline severity predicts risk well, even when treatment effect is constant.

Causal tree

Split on X when there is evidence that treatment effects differ across the resulting groups.

Example: biomarker status changes the treated–control contrast.

Strong predictor of Y $\neq$ strong modifier of treatment effect

Ordinary Random Forest Is Not a Causal Forest

Ordinary random forest

- Goal: predict Y accurately.
- Splits reduce outcome prediction error.
- A strong prognostic variable may dominate even if it does not modify treatment effect.

Causal forest

- Goal: estimate heterogeneity in treatment effects.
- Splits seek regions with different treatment effects.
- Treatment and control information are compared within leaves.

Prediction heterogeneity ≠ treatment-effect heterogeneity

Causal Forest: Intuition

Grow many trees

Repeatedly partition covariate space into local neighborhoods.

Look for effect differences

Prefer splits that reveal treatment-effect heterogeneity rather than outcome levels alone.

Average locally

Combine many trees to estimate $\tau(x)$ for each covariate profile.

Forest = adaptive local comparison + aggregation across many trees

Causal Forest: The Main Idea

1. Local groups

Trees partition covariate space into neighborhoods of similar X.

2. Local causal contrasts

Within neighborhoods, use observed treatment and outcome information to estimate effects.

3. Ensemble

Average information across many trees to produce a CATE estimate at x.

Unlike an ordinary forest, splitting is designed for treatment-effect heterogeneity.

Honest Trees: Choosing Splits

Without sample separation

The same unusual outcomes can influence both the tree partition and the effect estimate.

This can exaggerate apparent heterogeneity.

Honest estimation

Use one subsample to choose splits and another to estimate treatment effects in the selected leaves.

Reduces adaptive estimation bias.

Honesty helps with adaptive bias; it does not remove confounding or lack of overlap.

What Causal Forests Do and Do Not Guarantee

Useful features

- Learns complex interactions without listing every $T \times X$ term
- Pools information through many trees
- Produces a CATE prediction for a covariate profile
- Some implementations support statistical inference

Still required

- A defensible causal design and adjustment set
- Adequate overlap and sample size
- Valid outcome and treatment coding
- Diagnostics and honest uncertainty reporting

A sophisticated algorithm does not identify a causal effect by itself.

R: Fit a Causal Forest with the LaLonde Data

R code

```
library(MatchIt)
library(grf)
data("lalonde", package = "MatchIt")
d <- lalonde
X <- model.matrix(
  ~ age + educ + race + married + nodegree + re74 + re75,
  data = d)[, -1, drop = FALSE]

cf <- causal_forest(
  X, Y = d$re78, W = d$treat, seed = 123
)
```

Explain in class

X: baseline covariate matrix

Y: 1978 earnings

W: training indicator

The forest estimates effects conditionally on X.

Requires package grf.

What Does Variation in Predicted CATEs Mean?

Possible signal

The causal effect really differs across baseline covariate profiles.

Possible noise

Small subgroups, model instability, or finite-sample variability create apparent differences.

Needed next

Calibration, uncertainty, external or honest validation, and overlap checks.

Prediction variability is a hypothesis about HTE, not a completed validation.

Can We Trust the Heterogeneity?

Calibration

Are estimated effects systematically too small or too large?

Ranking

Do high-predicted-benefit groups show larger treatment effects in independent data?

Support + uncertainty

Are estimates based on adequate treatment overlap and enough information?

Evaluate the CATE model, not only its ability to predict the observed outcome.

Why In-Sample HTE Patterns Can Mislead

Discover + evaluate on the same data

The model can search for spurious groups.

Examining the same observations again may make those groups look more convincing.

Use independent evidence

Train on one set and evaluate HTE on held-out data.

Cross-fitting can use each observation for evaluation without fitting its prediction on itself.

Exploratory subgroup discovery should be distinguished from confirmatory evidence.

Cross-Fitting: The Basic Workflow

Fold A

Fit models using data outside Fold A.

Predict CATE for Fold A.

Fold B

Fit models using data outside Fold B.

Predict CATE for Fold B.

Repeat

Every evaluated person receives a prediction from a model that did not train on their outcome.

Cross-fitting limits overfitting; it does not fix unmeasured confounding.

CATE Calibration: Does the Magnitude Make Sense?

Example

Model predicts effects near −2 for low-risk patients and −10 for high-risk patients.

In independent evaluation data, do estimated treatment contrasts align with those magnitudes?

What calibration checks

Whether stronger predicted effects correspond to stronger causal contrasts.

Needs an appropriate causal estimator and uncertainty, not raw outcome comparisons.

CATE calibration ≠ ordinary outcome-prediction calibration.

Ranking by Predicted Benefit

First

From training data, learn $\hat{\tau}(x)$.

Second

In held-out data, group units into low, medium, and high predicted benefit.

Third

Estimate each group's causal effect, with adjustment appropriate to the design.

Do not compare untreated outcomes with treated outcomes without a credible design.

Local Overlap: Some CATEs Are Harder to Learn

$$e(x) = P(T = 1 \mid X = x)$$

At $e(x) \approx 0.5$

Treated and control observations are both reasonably represented nearby.

At $e(x) \approx 0.01$ or 0.99

One treatment arm is rare; local counterfactual comparisons are weakly supported.

A forest cannot recover an unobserved treatment arm from modeling alone.

Uncertainty: Estimated CATE Not an Individual Truth

Report

- Predicted treatment effects at relevant X
- Appropriate standard errors / intervals
- Uncertainty in subgroup differences
- Sample size and overlap diagnostics

Avoid

- Treating each $\hat{\tau}(x)$ as a known individual effect
- Calling a visually large difference significant
- Extrapolating to unsupported X regions
- Treating noisy rankings as clinical fact

If intervals are wide, say the data are not informative enough about heterogeneity.

R: Useful Causal-Forest Diagnostics

R code

```
# Calibration diagnostic within grf
test_calibration(cf)

# Forest's estimated average treatment effect
average_treatment_effect(cf)

# Treatment propensity estimated by forest
summary(cf$W.hat)

# Inspect CATE predictions
quantile(predict(cf)$predictions,
         probs = c(.05, .25, .5, .75, .95))
```

Important qualification

test_calibration() is a model diagnostic—not external validation.

For a real HTE validation exercise, use independent/held-out comparisons and appropriate inference.

What Should Responsible HTE Results Include?

Design first

- Population and estimand
- Treatment/outcome definitions
- Measured confounders
- Overlap assumptions

Method + diagnostics

- Model specification
- Sample splitting / honesty
- CATE plots and intervals
- Calibration and stability

Interpretation

- Magnitude and direction
- Clinically/policy meaningful strata
- Exploratory vs confirmatory
- Unmeasured confounding caveat

Explain who benefits, how much, and how confidently—not just which algorithm won.

Final HTE Takeaways and the Next Causal Challenge

What we can now do

- Model treatment × X interactions
- Estimate CATE with S/T/X learners
- Estimate flexible CATE with causal forests
- Evaluate heterogeneity rather than assuming it

What remains unresolved

All of these approaches require a defensible identification strategy.

What if treatment assignment depends on an important confounder U that was never measured?

Next: Unobserved Confounding and Sensitivity Analysis